

Maintenance Schedule

Diesel-electric drilling unit with Series 4000 engine

TPDSZ12V4000-2A0

12 V 4000 G23

Application group 3B

MS50125/01E



Power. Passion. Partnership.

System designation	Engine model	Hz.	kW	1/min	Application group
TPDSZ12V4000-2A0	12V4000G23	50 Hz.	1420 kW	1500 1/min	3B, Continuous operation, variable load, ICXN

Table 1: Applicability

1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Components relevant to emissions, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU, or equivalent components, are used.

The MTU maintenance system is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).

QL3: Maintenance work that requires partial disassembly of the product.

QL4: Maintenance work that requires complete disassembly of the product.

Supplementary information on maintenance and preservation:

Change intervals for fluids and lubricants and details on preservation can be obtained in the relevant specifications of the component manufacturers and MTU Fluids and Lubricants Specifications. The latest version is available at: <http://www.mtu-online.com>.

Manufacturer's specifications shall apply to the maintenance of all components not listed in this maintenance schedule.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

Application Group

3B Continuous operation, variable load, ICXN

Load factor

Load factor 40%

1.2 Maintenance schedule matrix

0-14,500 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W1024	1 w																														
W0500	-	X																													
W0501	-	X																													
W0502	-	X																													
W0503	-	X																													
W0505	-	X																													
W0506	-	X																													
W0507	-	X																													
W0508	-	X																													
W0525	-	X																													
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1008	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1247	3 m			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a					X				X				X				X				X				X				X	
W1724	3 a					X				X				X				X				X				X				X	
W1207	18 a			X				X						X						X						X		X			
W1011	4 a													X												X					
W1250	5 a													X												X					
W1006	18 a													X												X					
W1046	9 a																									X					
W1041	9 a																									X					
W2000	9 a																									X					
W2001	9 a																									X					
W2002	9 a																									X					
W2003	9 a																									X					
W2004	9 a																									X					
W2006	9 a																									X					
W2007	9 a																									X					
W2009	9 a																									X					
W2010	9 a																									X					
W2018	9 a																									X					
W2062	9 a																									X					
W2070	9 a																									X					
W2110	9 a																									X					
W2155	9 a																									X					
W1063	18 a																									X					
W1058	18 a																									X					
W1251	5 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
w = weeks m = months a = years																															

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3010	18 a																														
W3018	18 a																														
W3042	18 a																														
W3063	18 a																														
W3073	18 a																														
W3083	18 a																														
W3103	18 a																														
W3108	18 a																														
W1544	18 a																														
Generator																															
W1258	1 w																														
W1557	-	X																													
W1445	-	X																													
W1406	6 m					X				X				X				X				X				X				X	
W1405	6 m					X				X				X				X				X				X				X	
W1558	6 m					X				X				X				X				X				X				X	
W1409	6 m					X				X				X				X				X				X				X	
W1221	3 a																	X													
W1559	3 a																	X													
W1233	3 a																	X													
W1560	3 a																	X													
W1561	3 a																														
W1780	18 a																														
Fuel prefilter																															
W1164	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Radiator																															
W1564	1 w																														
W1563	-	X																													
W1565	-			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1569	18 a																														
Preheater / Jacket water heater																															
W1450	18 a																									X					
W1451	18 a																									X					
W1452	18 a																									X					
W1477	18 a																									X					
Control system																															
W1571	-			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
Resilient coupling																															
W1573	-												X													X					
w = weeks m = months a = years																															